University of Florida, Department of Mathematics

Algebra First-Year Exam, August 2016, Part 1

Answer four problems. (If you turn in more, the first four will be graded.) Within reason, you may quote theorems as long as you state them clearly.

1. Let G be a cyclic group, and let H be a subgroup of G. Prove that H is cyclic.

2. Let G be a finite group, and suppose it is acting on the finite set Ω. Suppose N is a normal subgroup of G and N acts transitively on Ω. Let $\omega \in \Omega$, and let G_ω be the stabilizer in G of ω. Prove that $|G : N| = |G_\omega : G_\omega \cap N|$.

3. Let G be a finite group, let H be a subgroup of G, and let $n = |G : H|$ with $n > 1$.

(a) Prove that if G is simple then G is isomorphic to a subgroup of the symmetric group S_n.

(b) Give an example to show that, under the general assumptions of the question, if G is not simple it is possible that G is not isomorphic to any subgroup of S_n (and prove that your example has these properties).

4. Let p be a prime, and let G be a finite group such that each element of G has order a power of p. Prove that, if G is not trivial, then the center $Z(G)$ of G is not trivial. Deduce that G is nilpotent.

5. Let S_n be the symmetric group in $n \geq 1$ letters. Let $\sigma \in S_n$. Describe what is meant by the cycle type of σ. Prove that two elements $\sigma, \tau \in S_n$ are conjugate to each other in S_n if and only if they have the same cycle type.

6. Prove that a group of order 30 must have a normal subgroup of order 15.